

Patrick R. Oare

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EDUCATION

Ph.D. Student at the Massachusetts Institute of Technology (MIT)

August 2019 - September 2024

Department: Physics ([Center for Theoretical Physics](#))

GPA: 5.0/5.0

Thesis: [Beyond Color: Lattice Gauge Theory for Strongly-Coupled Physics](#)

Degree: [Ph.D. in Physics, Statistics, and Data Science](#)

Undergraduate Student at the University of California, Berkeley

August 2015 - May 2019

Degrees: B.A. Physics (Highest Honors), B.A. Mathematics (Highest Honors)

GPA: 3.96 / 4.0

Awards: Isidore Pomerantz Award, Physics Department; Dorothea Klumpke Roberts Prize, Mathematics Department; Phi Beta Kappa.

RESEARCH AND WORK EXPERIENCE

Research Associate at Brookhaven National Laboratory (BNL)

October 2024 - Present

Department: Physics (High Energy Theory)

You can find a more in-depth description of my research [on my website](#).

My primary research focus is in [lattice quantum chromodynamics \(QCD\)](#), which is a computational framework used to study the strong nuclear force via discretizing spacetime to evaluate the path integral numerically. Calculations use Markov Chain Monte Carlo to sample a high-dimensional probability distribution (order $\sim 12 \times 48^3 \times 96$ dimensions) and determine estimators for observables using resampling methods (bootstrap, jackknife). In a typical calculation, I use the [Grid software library](#) to compute these samples on a computer cluster, then use Python or Mathematica to process the statistical data.

My current projects include:

- **Acceleration of linear solvers in lattice QCD.** The fundamental bottleneck in most lattice QCD calculations is solving the Dirac equation $D\psi = b$, where D is a very high-dimensional sparse matrix. We are studying ways to accelerate solvers for this equation using algebraic multigrid methods.
- **Adjoint QCD in $d = 2$ spacetime dimensions (QCD₂).** Adjoint QCD is the theory of a Majorana fermion coupled to an $SU(N)$ gauge field in the adjoint representation in 2 spacetime dimensions. We are in the process of computing string tensions and the low-lying spectrum of QCD₂ using rational Hybrid Monte Carlo.
- **Reconstruction of spectral functions.** We are currently developing techniques to apply our spectral reconstruction methods to noisy Monte Carlo data (Publications [1-3](#)).

MIT Lattice QCD Group

August 2019 - September 2024

Doctoral Student

In graduate school I worked on the following topics:

- **Computing hadronic matrix elements of operators facilitating neutrinoless double β ($0\nu\beta\beta$) decay.** We computed the matrix elements for $0\nu\beta\beta$ decay in the $\pi^- \rightarrow \pi^+ e^- e^-$ and $n^0 n^0 \rightarrow p^+ p^+ e^- e^-$ decays using Hybrid Monte Carlo (HMC) and used χ^2 minimization to extrapolate the results to the continuum limit (Publications [4, 7](#)).
- **Reconstruction of spectral functions.** Spectral functions describe the interacting states in a quantum field theory, and are related to directly computable quantities by an ill-posed inverse problem. We developed a novel method for reconstructing spectral functions in lattice QCD based on Nevanlinna-Pick interpolation theory (Publication [6](#)).
- **Renormalization of local composite operators.** Lattice regularizations of quantum field theory must be connected to the physical continuum theory by renormalization. We computed the renormalization coefficients of a variety of composite operators using HMC and numerical integration (Publications [5, 8](#)).

Computational Imaging Lab

August 2016 - Spring 2019

Undergraduate Researcher

The Computational Imaging Lab studies Fourier Optics and its applications, along with other computational imaging techniques. I worked on reconstruction of a high-speed video from a single image in a rolling shutter camera. We used optical multiplexing to scatter light from the object plane into every pixel in the camera plane. This allowed us to invert a forward model and recover the original video that generated the image (Publication [9](#)).

PUBLICATIONS

Authorship in high-energy theory publications ([1-8](#)) is by alphabetical order.

1. R. Abbott, S. Fields, W. Jay, [P. Oare](#), M. Saccardi, **The Causal Bootstrap: Bounding Smeared Spectral Functions from Non-Perturbative Euclidean Data**, [arXiv:hep-lat/2605.20509](#) (accepted to Phys. Rev. D, September 2026).
2. R. Abbott, W. Jay, [P. Oare](#), **Moment problems and spectral functions**, [PoS LATTICE2025 \(2026\) 146](#).
3. R. Abbott, W. Jay, [P. Oare](#), **Moment problems and bounds for matrix-valued smeared spectral functions**, [arXiv:hep-lat/2508.01377](#) (in submission to Phys. Rev. D, September 2026).
4. W. Detmold, Z. Fu, A. Grebe, W. Jay, D. Murphy, [P. Oare](#), P. Shanahan, **Long-distance nuclear matrix elements for neutrinoless double-beta decay from lattice QCD**, [Phys. Rev. D 109 \(2024\) 11, 114514](#).
5. D. Hackett, [P. Oare](#), D. Pefkou, P. Shanahan, **Gravitational form factors of the pion from lattice QCD**, [Phys. Rev. D 108 \(2023\) 11, 114504](#).
6. T. Bergamaschi, W. Jay, [P. Oare](#), **Hadronic Structure, Conformal Maps, and Analytic Continuation**, [Phys. Rev. D 108 \(2023\) 7, 074516](#).
7. W. Detmold, W. Jay, D. Murphy, [P. Oare](#), P. Shanahan, **Neutrinoless Double Beta Decay from Lattice QCD: The Short-Distance $\pi^- \rightarrow \pi^+ e^- e^-$ Amplitude**, [Phys. Rev. D 107 \(2023\) 9, 094501](#).
8. W. Detmold, M. Illa, D. Murphy, [P. Oare](#), K. Orginos, P. Shanahan, M. Wagman, F. Winter, **Lattice QCD constraints on the parton distribution functions of ^3He** , [Phys. Rev. Lett. 126 \(2021\) 20, 202001](#).
9. N. Antipa*, [P. Oare](#)*, E. Bostan, R. Ng, L. Waller, **Video from Stills: Lensless Imaging with Rolling Shutter**, *2019 IEEE International Conference on Computational Photography (ICCP)*, University of Tokyo (May 2019). [Best Paper Award](#) at ICCP 2019.

TALKS AND PRESENTATIONS

1. Talk given at *The 43rd International Symposium on Lattice Field Theory*, **Spectral Properties of non-Hermitian Dirac Operators from the Krylov-Schur Algorithm**. College Park, MD (July 2026).
2. Talk given at *The 41st Colorado Conference on Iterative and Multigrid Methods*, **The Wilson-Dirac Operator and its Spectrum**. Boulder, CO (June 2026).
3. Talk given at the *Columbia Theory Seminar*, **Non-Hermitian Eigensolvers for Lattice Gauge Theory**. New York, NY (May 2026).
4. Talk given at the *2024 Simons Collaboration Meeting on Confinement and QCD Strings*, **Low-Lying Spectrum of Two-Dimensional Adjoint QCD from the Lattice**. Princeton, NJ (November 2024).
5. Talk given at *The 41st International Symposium on Lattice Field Theory*, **Low-Lying Spectrum of Two-Dimensional Adjoint QCD from the Lattice**. Liverpool, United Kingdom (July 2024).
6. Talk given at *The 40th International Symposium on Lattice Field Theory*, **Neutrinoless Double Beta Decay from Lattice QCD: The $n^0 n^0 \rightarrow p^+ p^+ e^- e^-$ Amplitude**. Fermilab, IL (August 2023).
7. Talk given at *Prospects in Theoretical Physics 2023: "Understanding Confinement"*, **2d Adjoint QCD on the Lattice**. Princeton, NJ (July 2023).
8. Talk given at *The 39th International Symposium on Lattice Field Theory*, **Neutrinoless Double Beta Decay from Lattice QCD: The Short-Distance $\pi^- \rightarrow \pi^+ e^- e^-$ Amplitude**. Bonn, Germany (August 2022).
9. Seminar presented to the Berkeley Center for Computational Imaging, **Video from Stills: Using Rolling Shutter and Spatial Multiplexing to Image Fast-Moving Scenes**. Berkeley, CA (September 2018).

SKILLS

- Programming Languages:
 - Advanced Proficiency: Bash, C++, Java, Julia, Lua, Mathematica, Python, XML.
 - Basic Proficiency: Arduino, LabVIEW, MATLAB, PHP, Scheme, SQL.
- Distributed and Parallel Computing: [CUDA](#), [OpenMP](#), [MPI](#), [SLURM](#).
- High-performance computing clusters: [ALCF Aurora](#), [OLCF Frontier](#), [BNL SDCC](#).
- Lattice QCD Libraries: [Chroma](#), [Grid](#), [QLUA](#), [QUDA](#).
- **Math proficiencies**: Multigrid & finite-element methods, iterative solvers, statistics, probability & measure theory, convex optimization, linear & abstract algebra, real & complex analysis, algebraic topology, differential geometry.
- **Physics proficiencies**: quantum field theory, the Standard Model, classical mechanics, electricity & magnetism, quantum mechanics, special & general relativity, and statistical mechanics.

TEACHING EXPERIENCE

Relativistic Quantum Field Theory I (8.323)

Spring 2021, 2022

Recitation and grading TA for graduate-level Quantum Field Theory I.

- Developed a curriculum for weekly recitation. My class notes can be found [on my website](#).
- Average rating from student evaluations: 6.8/7.0 (Spring 2021), 6.9/7.0 (Spring 2022).

OUTREACH, COMMUNITY SERVICE, AND DEPARTMENT INVOLVEMENT

Lectures on Tap speaker

Fall 2026 - Present

[Lectures on Tap](#) is an organization that schedules public lectures from experts in bars. In September 2026 I gave a lecture titled “How We Discovered the Building Blocks of Matter”.

BNL High-Energy Theory Seminar

Fall 2025 - Fall 2026

As a co-organizer for the High-Energy Theory Seminar, I was responsible for inviting and coordinating with speakers and organizing speaker dinners.

Teaching Development Fellow (TDF), Physics

Fall 2021 - Spring 2023

The [TDF network](#) is a professional development and leadership opportunity for graduate students who are interested in improving their teaching while promoting pedagogical development and discussion among their peers.

As physics TDF, I was the liaison between MIT’s [Teaching and Learning Lab](#) and the Teaching Assistants (TAs) in the Physics department. I ran workshops aimed at increasing the pedagogical knowledge of these TAs, and interfaced with department leaders to improve the TA experience.

Physics Graduate Student Council (PGSC)

Fall 2020 - Spring 2023

Social Chair, Athletic Chair

The [PGSC social chairs](#) are responsible for organizing social gatherings for the graduate students in the department. As athletic chair, I was responsible for interfacing between MIT Intramural Sports and the physics graduate students, and organizing graduate students into teams.

CTP Graduate Student Lunch Club

Fall 2019 - Spring 2020

As an organizer of the [Graduate Student Lunch Club](#), I was responsible for recruiting graduate student and postdoc speakers for our weekly seminars, coordinating with the speakers, and promoting these talks throughout the CTP.

Berkeley Engineers and Mentors (BEAM, <https://beam.berkeley.edu/>)

Fall 2015 - Spring 2019

President ('18 - '19), Site Leader and Curriculum Creator ('16 - '19)

BEAM is a group of UC Berkeley students dedicated to furthering STEM education among the local community. We work with local schools to bring science lessons to children. As President, I facilitated a class (ChemE 98/198) at UC Berkeley with over 100 students, where we taught our mentors pedagogical and classroom management techniques.

INTERESTS AND ACTIVITIES

Ice Hockey; Rock Climbing; Obstacle Course Racing; Tennis; Snowboarding; World History & Geopolitics